

Project_1 Cell library

CE7325 AVLSI

LAMIN JAMMEH (DAL852207) & MARK IVEY (MXI240020)

Abstract:

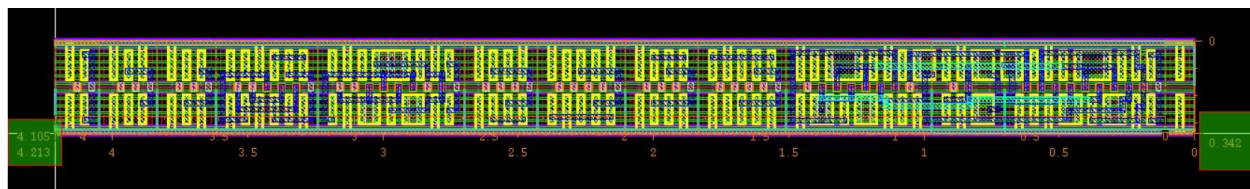
The cells for this Project were design using FinFET technology. The fin width, fin pitch are kept universal for all the cells. The gate pitch and cell offset was also kept the same.

Design specs:

- Fin width
- Fin Pitch
- Gate Pitch
- Cell offset

Combine_cell:

- Cell Height (VDD to GND) = 0.342um
- Cell width (with tap cell) = 4.213um
- Cell width (without tap cell) = 4.105um

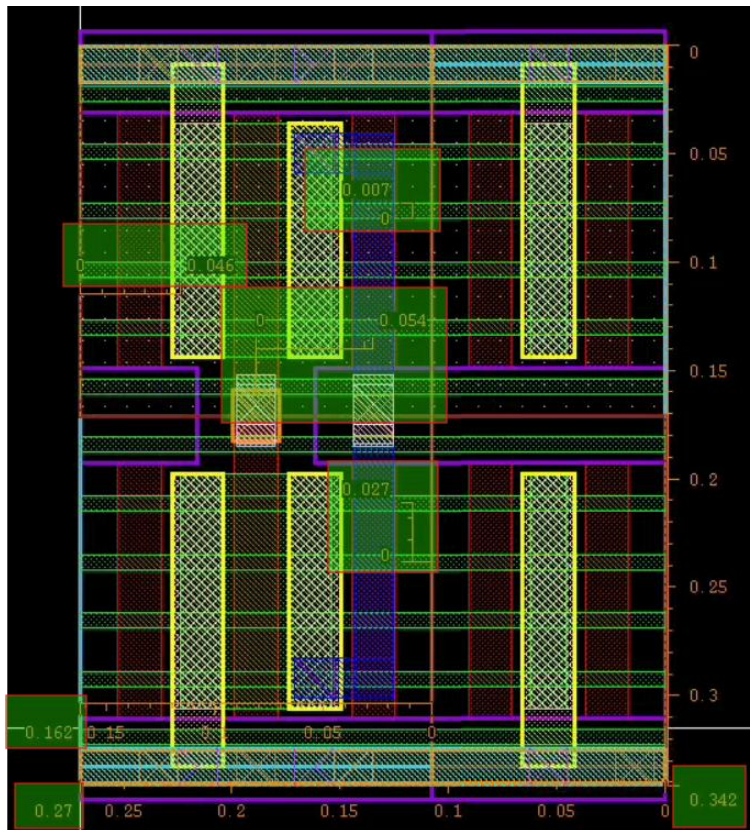


INV:

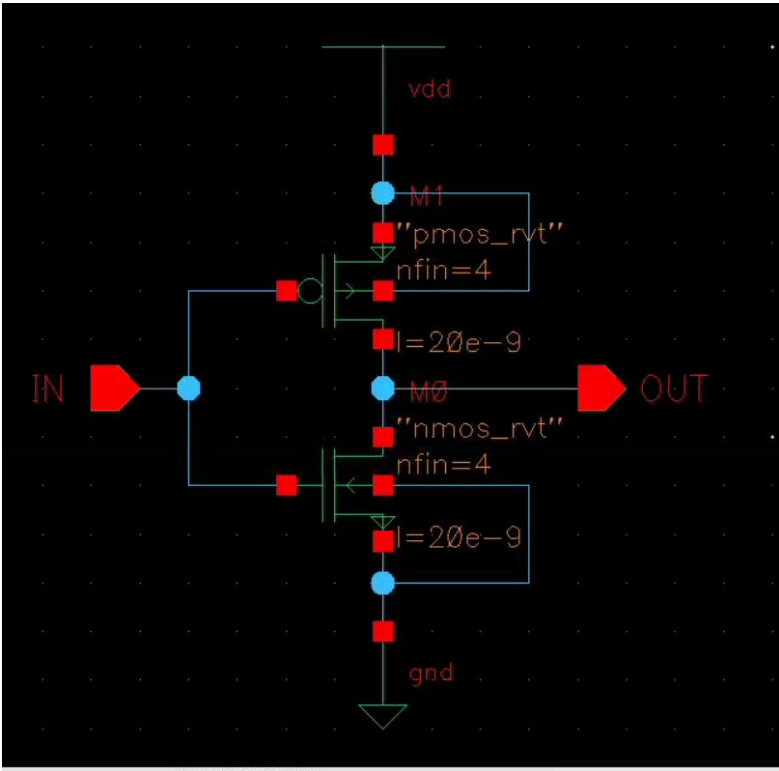
Layout design

The following Specs are used for the design

- Cell Height (VDD to GND) = 0.342um
- Cell width (with tap cell) = 0.27um
- Cell width (without tap cell) = 0.162um
- Cell offset = 0.046um
- Fin Pitch = 0.027um
- Fin width = 0.007um
- Gate Pitch (Pin to Pin Distance) = 0.054um



Schematic:



LVS Result:

CALIBRE SYSTEM ##
LVS REPORT ##
#####

REPORT FILE NAME: INV.lvs.report
LAYOUT NAME: INV.sp ('INV')
SOURCE NAME: INV.src.net ('INV')

RULE FILE: _Calibre_LVS_rules_
RULE FILE TITLE: LVS Rule File for ASAP7 PDK
CREATION TIME: Mon Mar 24 01:41:09 2025
CURRENT DIRECTORY: /home/eng/i/dal852207/asap7/LVS_check_INV
USER NAME: dal852207
CALIBRE VERSION: v2024.4_12.9 Tue Oct 1 20:34:12 PDT 2024

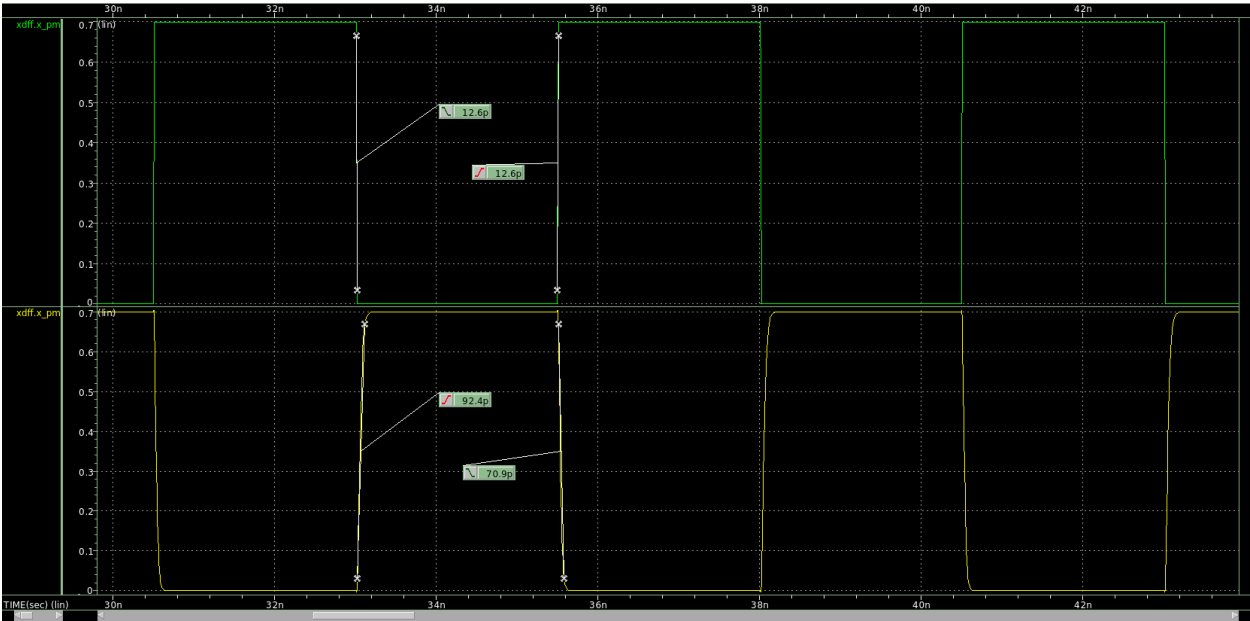
OVERALL COMPARISON RESULTS

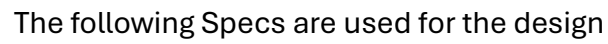
#

CORRECT #
#

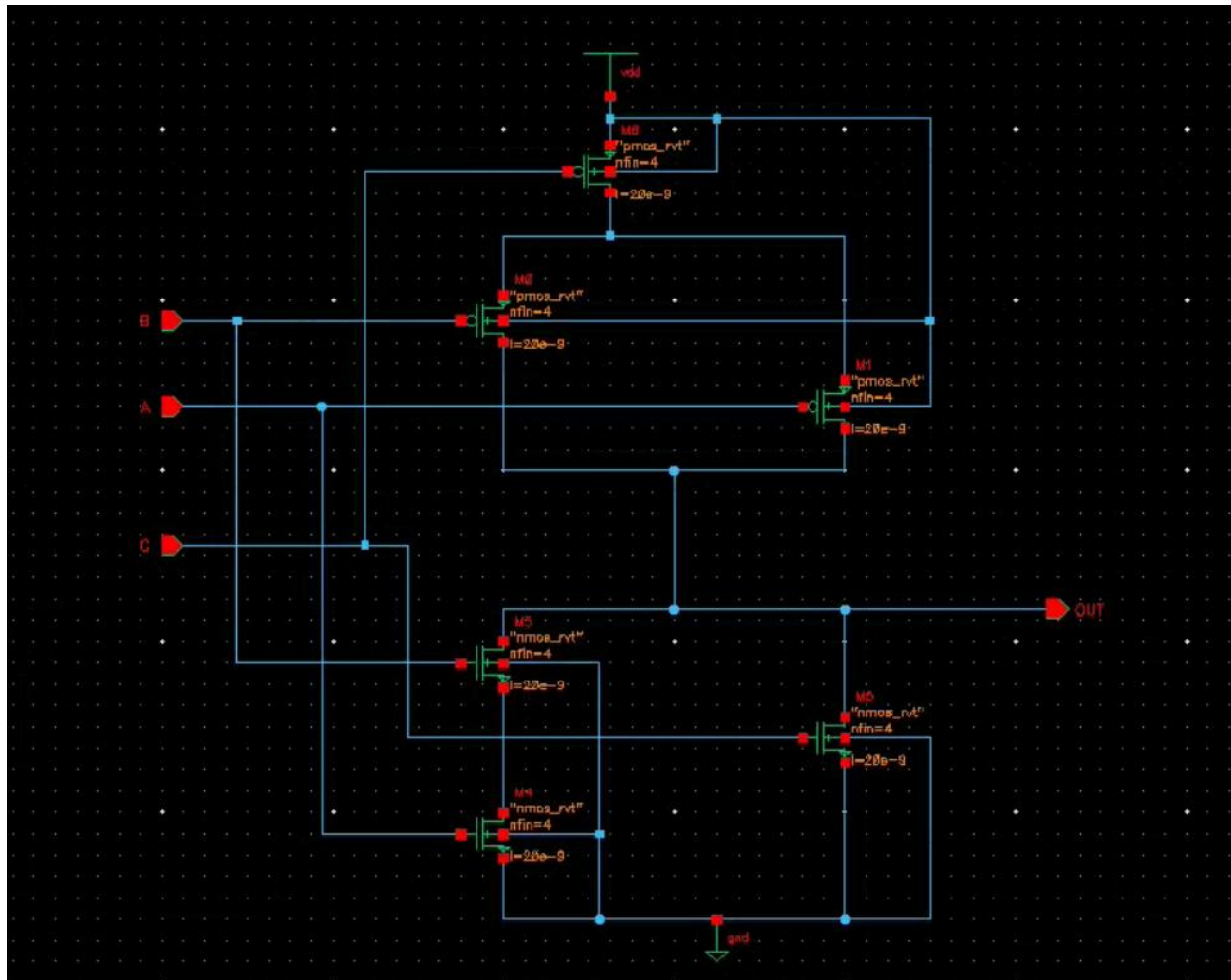
* *
|
_ _

Simulation Waveform:





Schematic:



```
#####
##                                ##
##      C A L I B R E   S Y S T E M      ##
##                                ##
##      L V S   R E P O R T              ##
##                                ##
#####

REPORT FILE NAME:      AOI21.lvs.report
LAYOUT NAME:          AOI21.sp ('AOI21')
SOURCE NAME:          AOI21.src.net ('AOI21')
RULE FILE:            _Calibre_LVS_rules_
RULE FILE TITLE:      LVS Rule File for ASAP7 PDK
CREATION TIME:        Tue Mar 25 23:54:30 2025
CURRENT DIRECTORY:    /home/eng/m/mxi240020/7325/asap7/Std_cells/AOI21/LVS
USER NAME:            mxi240020
CALIBRE VERSION:      v2024.4_12.9    Tue Oct 1 20:34:12 PDT 2024
```

```

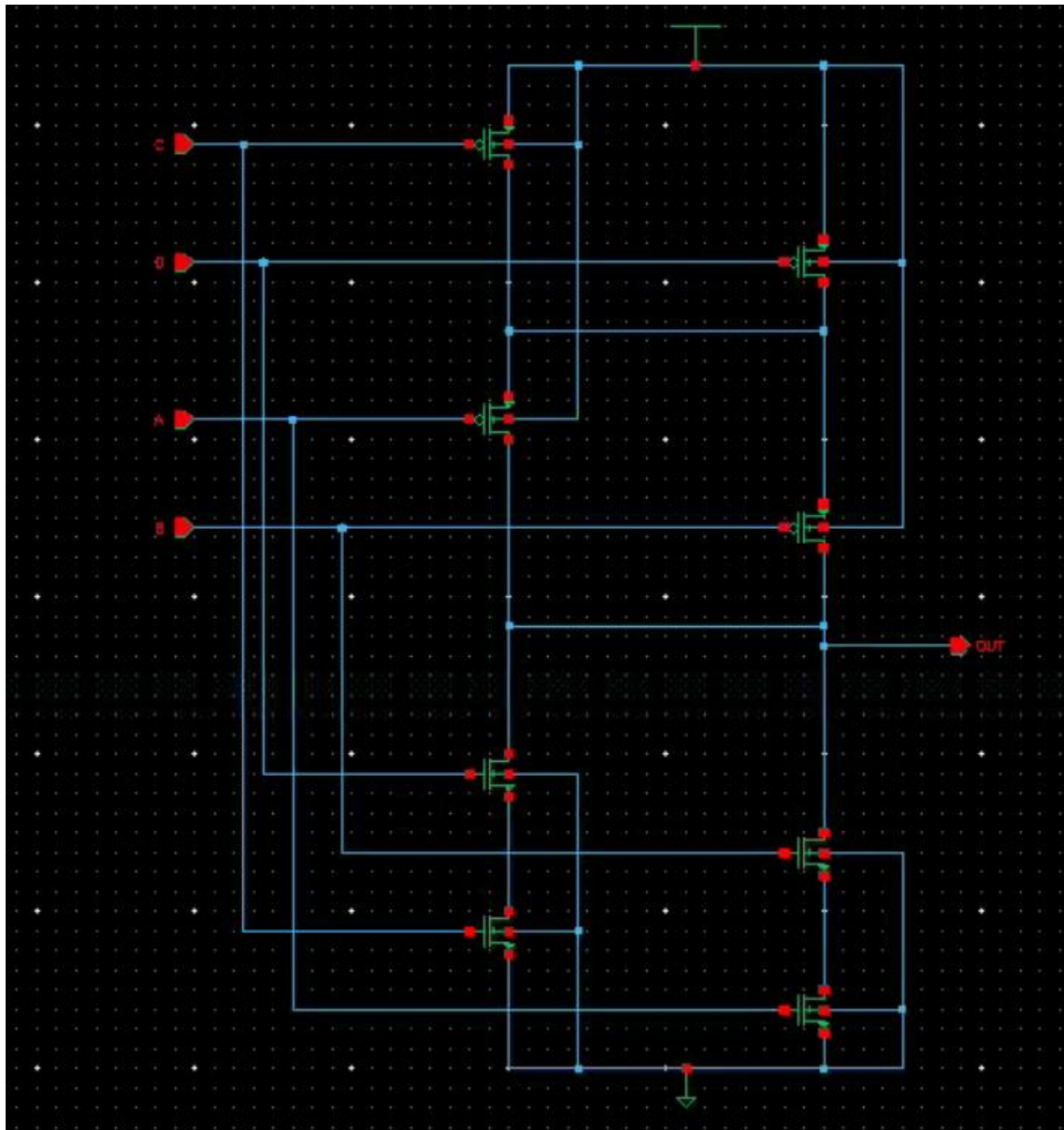
# #####
# #
# # CORRECT #
# #
# #####

```

The timing diagram displays four signals over a time interval from 20n to 40n seconds. The signals are labeled as $xdiff_x_pn$ and show different levels and transitions. Annotations indicate specific transition times: 12.6p and 46p.



Schematic:



LVS Result:

```
#####
##                                     ##
##          C A L I B R E   S Y S T E M          ##
##                                     ##
##          L V S   R E P O R T                   ##
##                                     ##
#####
```

```
REPORT FILE NAME:      AOI22.lvs.report
LAYOUT NAME:          AOI22.sp ('AOI22')
SOURCE NAME:           AOI22.src.net ('AOI22')
RULE FILE:             _Calibre_LVS_rules_
RULE FILE TITLE:       LVS Rule File for ASAP7 PDK
CREATION TIME:         Wed Mar 26 00:04:11 2025
CURRENT DIRECTORY:     /home/eng/m/mxi240020/7325/asap7/Std_cells/AOI22/LVS
USER NAME:             mxi240020
CALIBRE VERSION:       v2024.4.12.9    Tue Oct 1 20:34:12 PDT 2024
```

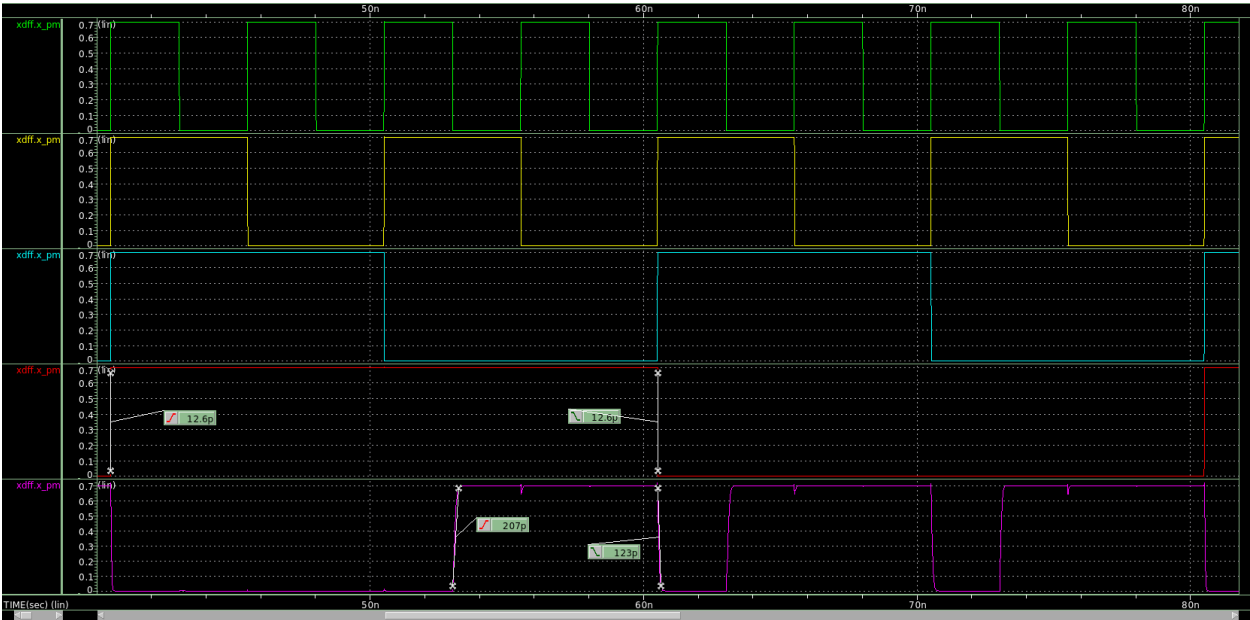
OVERALL COMPARISON RESULTS

```

#
#
# #
# #
#
#####
#
# CORRECT #
#
#####

```

Simulation Waveform:

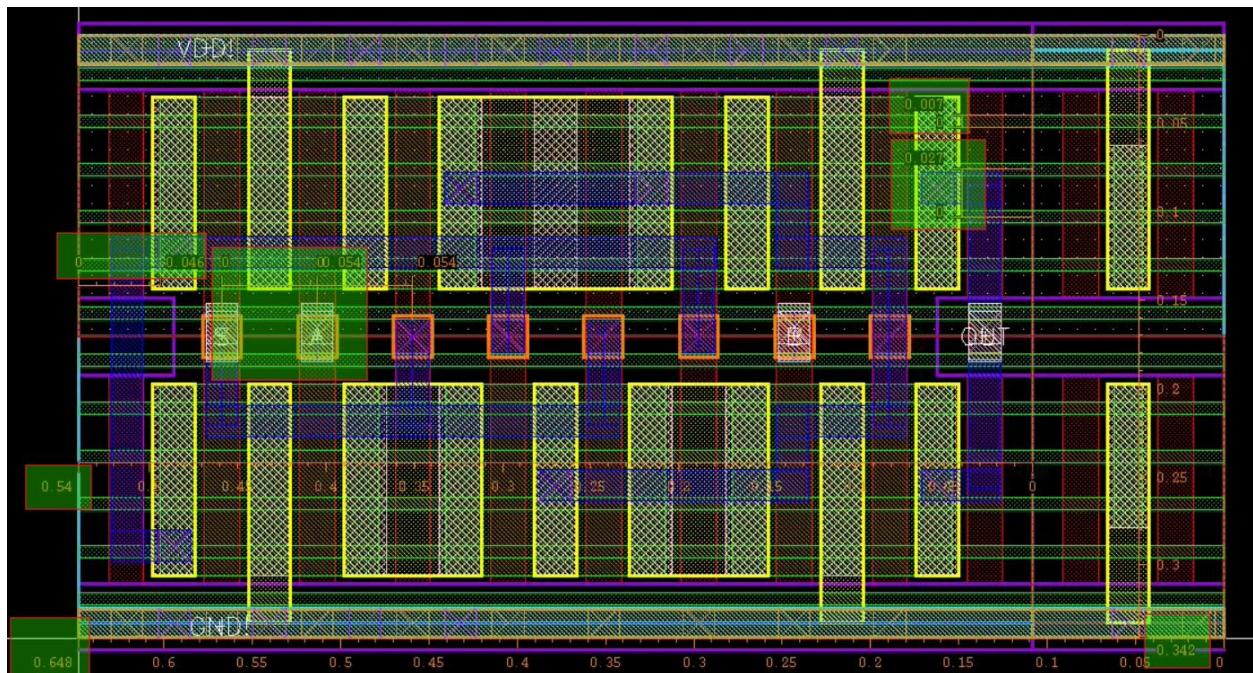


MUX2:1:

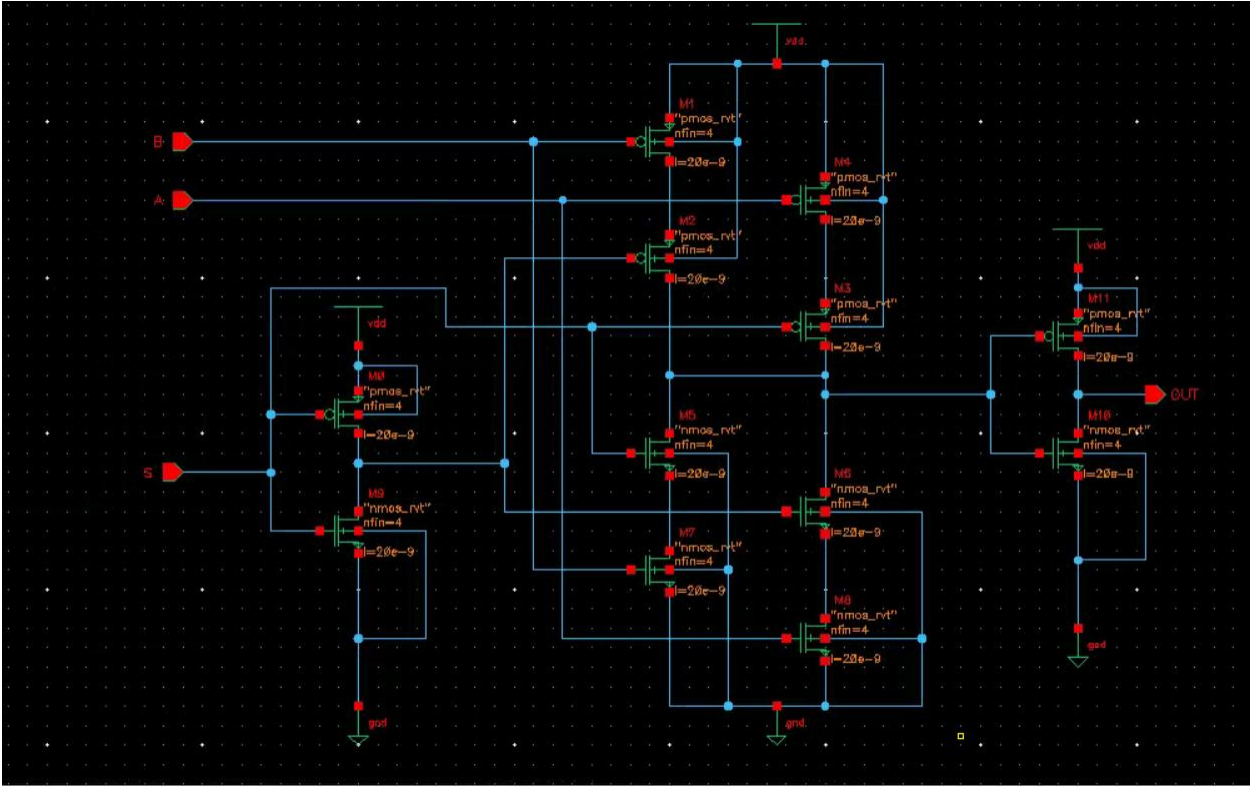
Layout design

The following Specs are used for the design

- Cell Height (VDD to GND) = 0.342um
- Cell width (with tap cell) = 0.648um
- Cell width (without tap cell) = 0.54um
- Cell offset = 0.046um
- Fin Pitch = 0.027um
- Fin width = 0.007um
- Gate Pitch (Pin to Pin Distance) = 0.054um



Schematic:



LVS Result:

```
#####
##          CALIBRE SYSTEM          ##
##                                     ##
##          LVS REPORT                ##
##                                     ##
#####

REPORT FILE NAME:  MUX2_1.lvs.report
LAYOUT NAME:      MUX2_1.sp ('MUX2_1')
SOURCE NAME:      MUX2_1.src.net ('MUX2_1')
RULE FILE:        _Calibre_LVS_rules_
RULE FILE TITLE:  LVS Rule File for ASAP7 PDK
CREATION TIME:    Tue Mar 25 22:47:41 2025
CURRENT DIRECTORY: /home/eng/m/mxi240020/7325/asap7/Std_cells/MUX2_1/LVS
USER NAME:        mxi240020
CALIBRE VERSION:  v2024.4_12.9    Tue Oct 1 20:34:12 PDT 2024
```

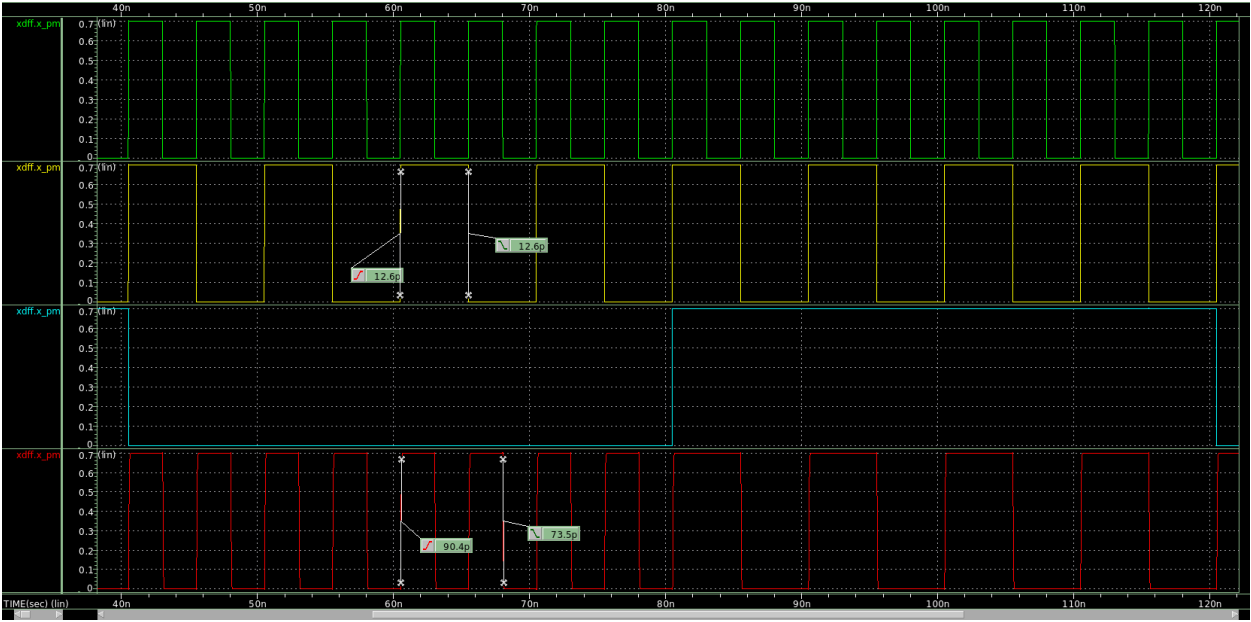
OVERALL COMPARISON RESULTS

#

CORRECT #
#####

* *
|
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Simulation Waveform:

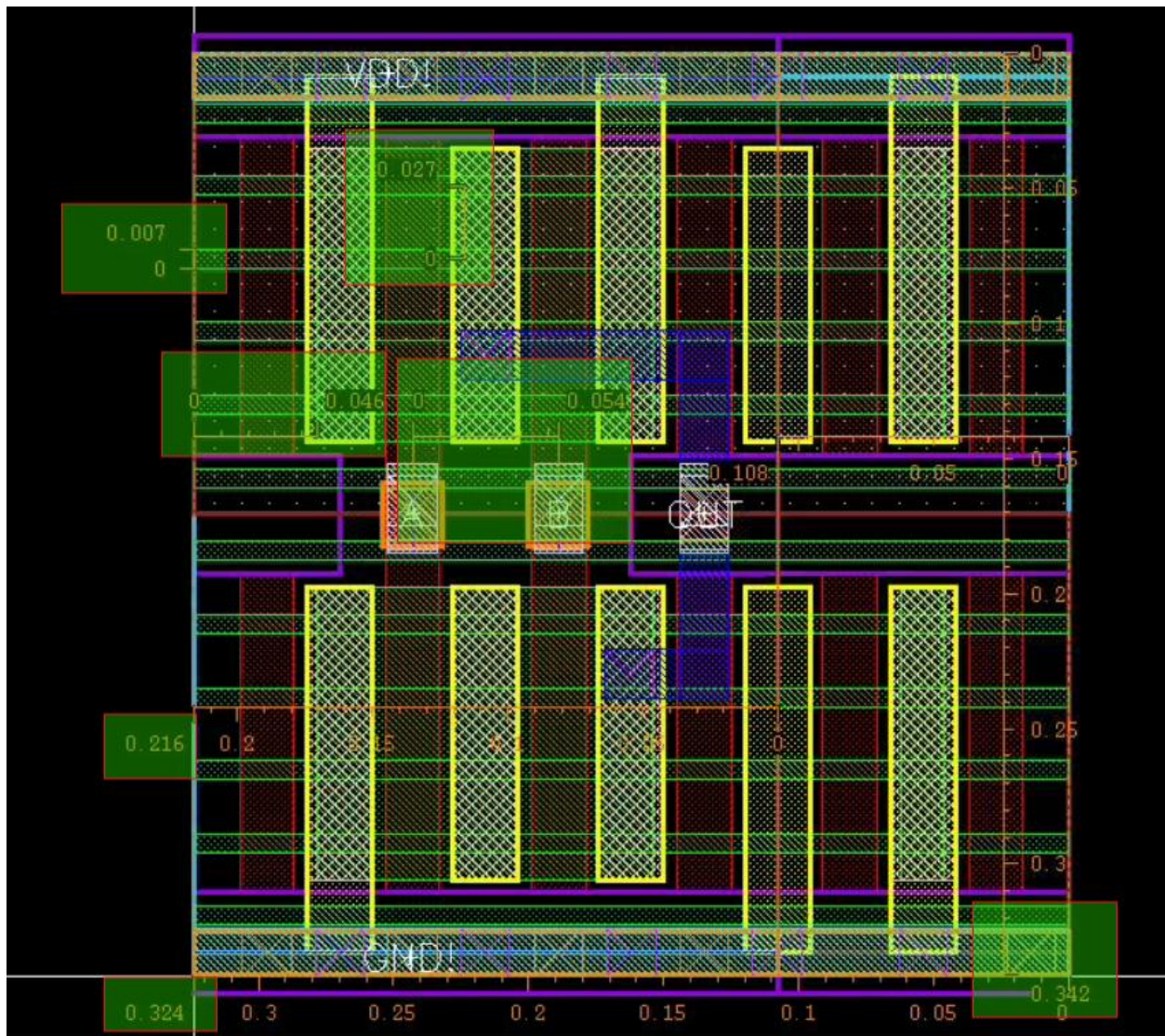


NAND2:

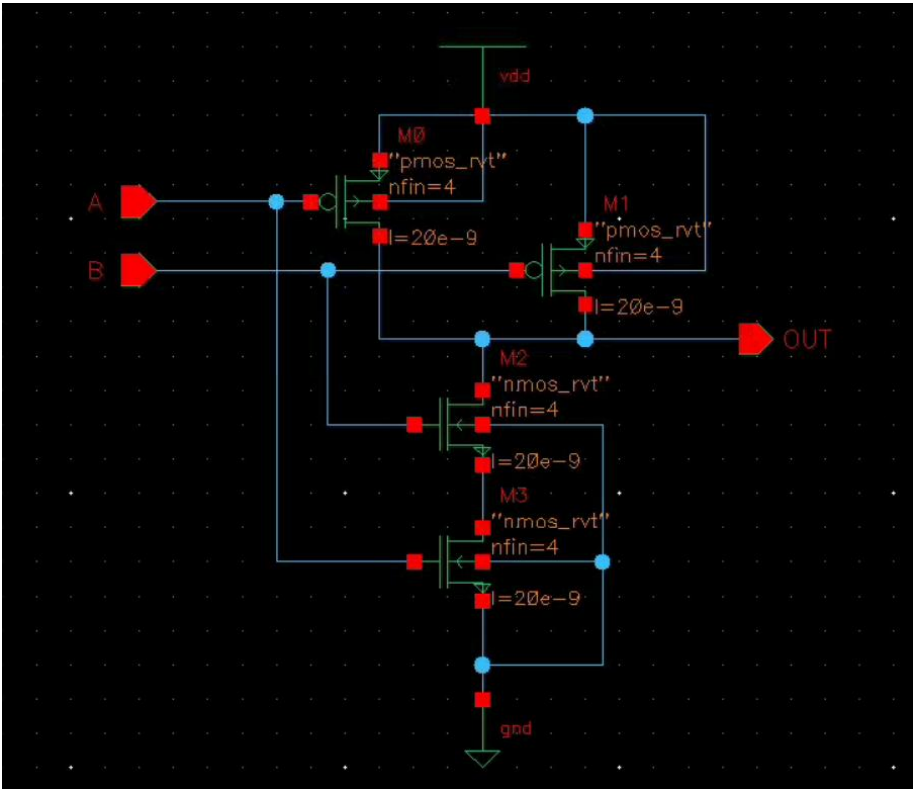
Layout design

The following Specs are used for the design

- Cell Height (VDD to GND) = $0.342\mu\text{m}$
- Cell width (with tap cell) = $0.324\mu\text{m}$
- Cell width (without tap cell) = $0.216\mu\text{m}$
- Cell offset = $0.046\mu\text{m}$
- Fin Pitch = $0.027\mu\text{m}$
- Fin width = $0.007\mu\text{m}$
- Gate Pitch (Pin to Pin Distance) = $0.054\mu\text{m}$



Schematic:



LVS Result:

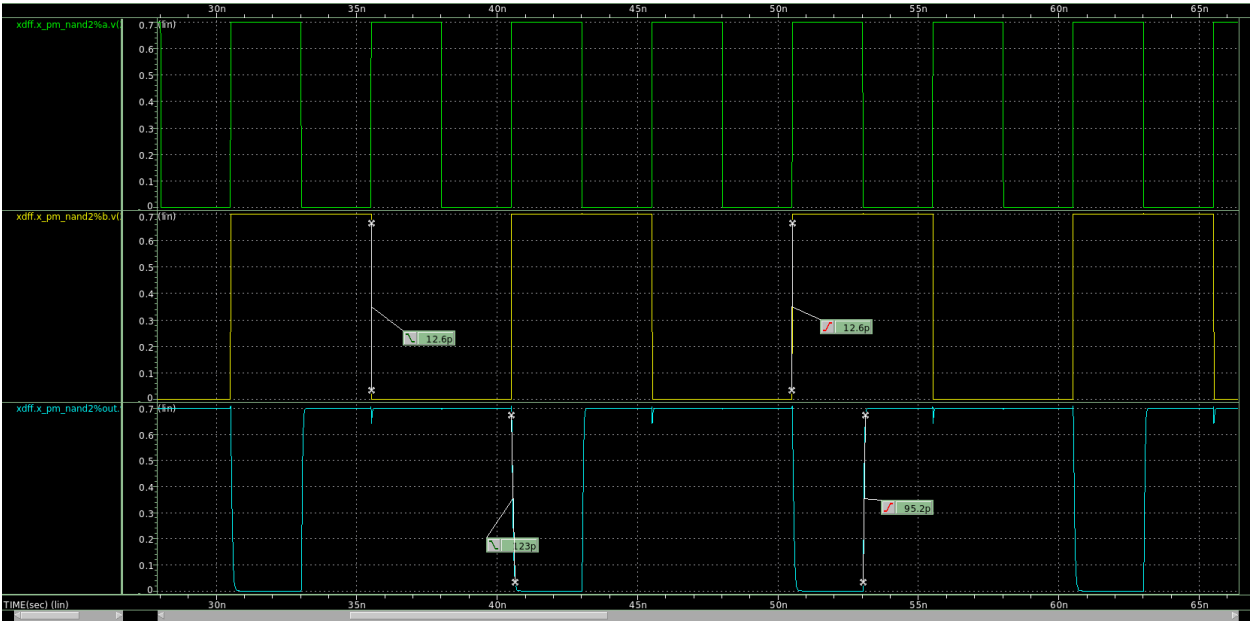
```
#####
##          CALIBRE SYSTEM          ##
##                                     ##
##          LVS REPORT                ##
##                                     ##
#####

REPORT FILE NAME:  NAND2.lvs.report
LAYOUT NAME:      NAND2.sp ('NAND2')
SOURCE NAME:      NAND2.src.net ('NAND2')
RULE FILE:        _Calibre_LVS_rules_
RULE FILE TITLE:  LVS Rule File for ASAP7 PDK
CREATION TIME:    Wed Mar 26 00:08:50 2025
CURRENT DIRECTORY: /home/eng/m/mxi240020/7325/asap7/Std_cells/NAND2/LVS
USER NAME:        mxi240020
CALIBRE VERSION:  v2024.4_12.9    Tue Oct 1 20:34:12 PDT 2024
```

```
OVERALL COMPARISON RESULTS

#          #####
#          #          *  *
#  #      #  CORRECT  #
#  #      #          #
#          #####
```

Simulation Waveform:

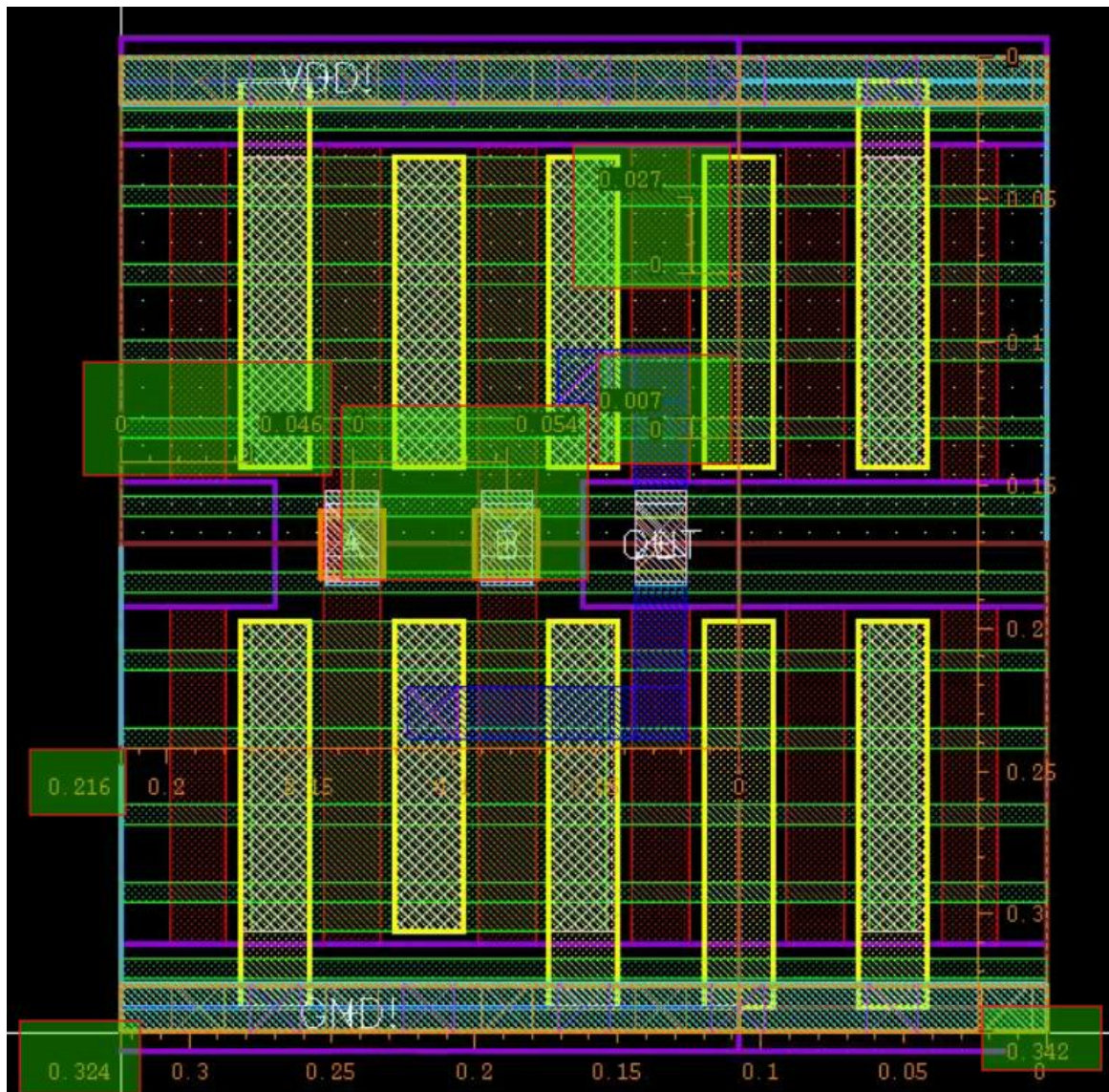


NOR2:

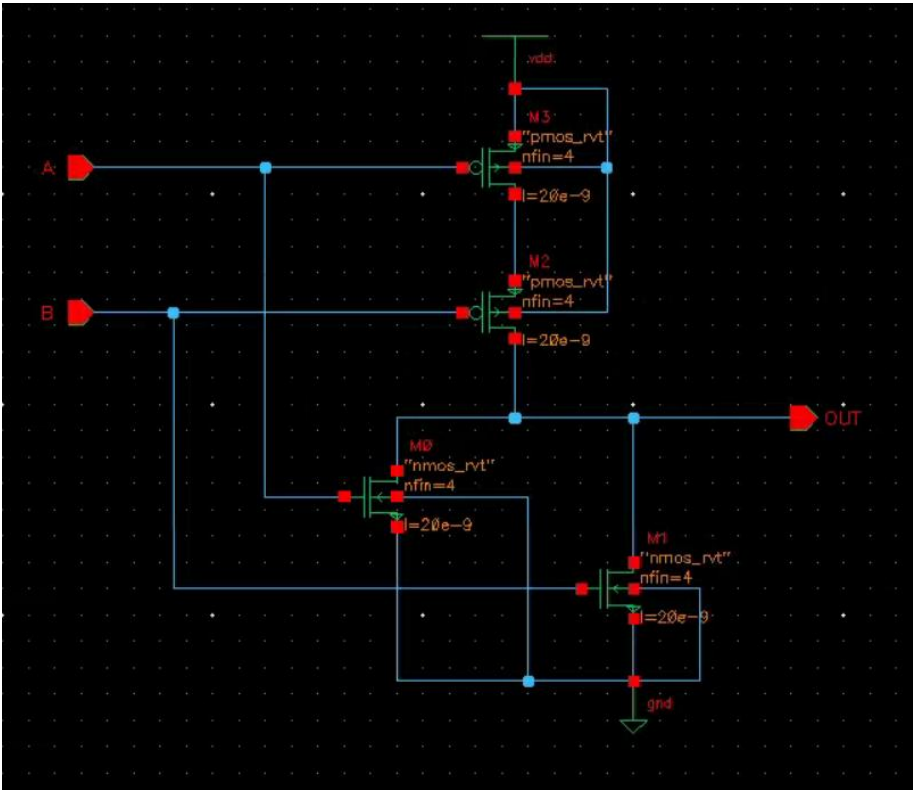
Layout design

The following Specs are used for the design

- Cell Height (VDD to GND) = 0.342um
- Cell width (with tap cell) = 0.324um
- Cell width (without tap cell) = 0.216um
- Cell offset = 0.046um
- Fin Pitch = 0.027um
- Fin width = 0.007um
- Gate Pitch (Pin to Pin Distance) = 0.054um



Schematic:



LVS Result:

CALIBRE SYSTEM ##
LVS REPORT ##
#####

REPORT FILE NAME: NOR2.lvs.report
LAYOUT NAME: NOR2.sp ('NOR2')
SOURCE NAME: NOR2.src.net ('NOR2')

RULE FILE: _Calibre_LVS_rules_
RULE FILE TITLE: LVS Rule File for ASAP7 PDK
CREATION TIME: Wed Mar 26 00:15:24 2025
CURRENT DIRECTORY: /home/eng/m/mxi240020/7325/asap7/Std_cells/NOR2/LVS
USER NAME: mxi240020
CALIBRE VERSION: v2024.4_12.9 Tue Oct 1 20:34:12 PDT 2024

OVERALL COMPARISON RESULTS

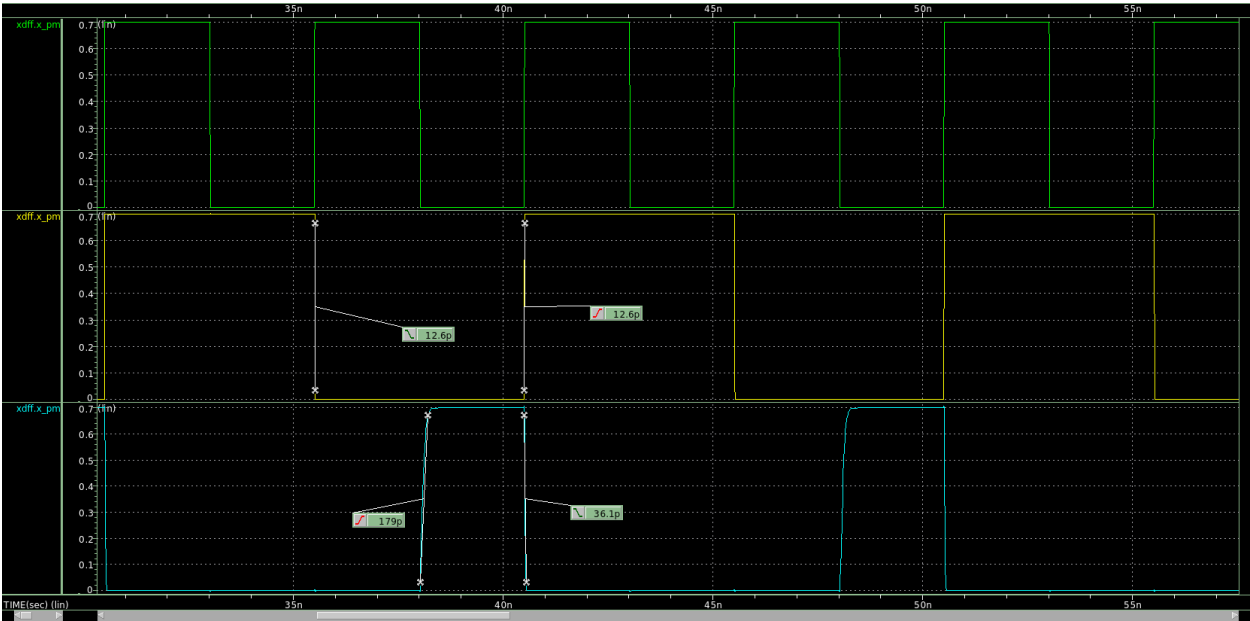
#

CORRECT #

#####

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Simulation Waveform:

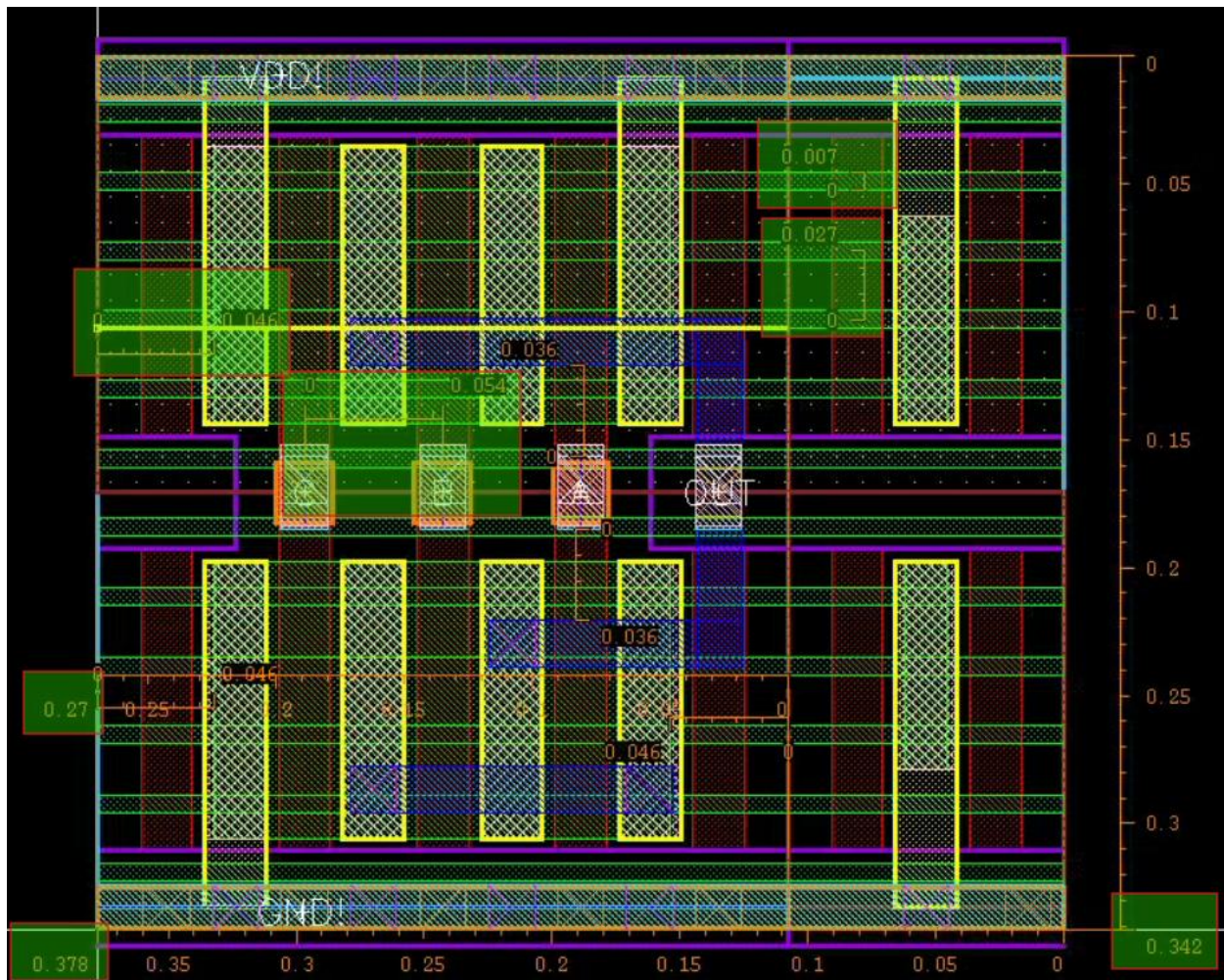


OAI21:

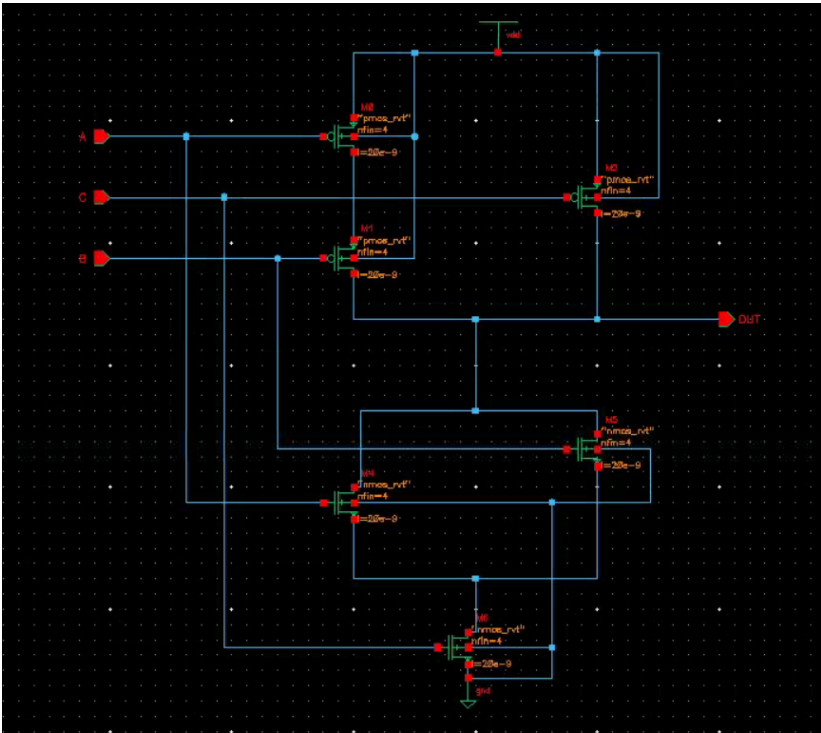
Layout design

The following Specs are used for the design

- Cell Height (VDD to GND) = 0.342um
- Cell width (with tap cell) = 0.378um
- Cell width (without tap cell) = 0.27um
- Cell offset = 0.046um
- Fin Pitch = 0.027um
- Fin width = 0.007um
- Gate Pitch (Pin to Pin Distance) = 0.054um



Schematic:



LVS Result:

```
#####
##          CALIBRE SYSTEM          ##
##          LVS REPORT              ##
##          #####
```

```
REPORT FILE NAME:  OAI21.lvs.report
LAYOUT NAME:      OAI21.sp ('OAI21')
SOURCE NAME:      OAI21.src.net ('OAI21')
RULE FILE:        _Calibre_LVS_rules_
RULE FILE TITLE:   LVS Rule File for ASAP7 PDK
CREATION TIME:     Wed Mar 26 00:19:55 2025
CURRENT DIRECTORY: /home/eng/m/mxi240020/7325/asap7/Std_cells/OAI21/LVS
USER NAME:         mxi240020
CALIBRE VERSION:   v2024.4_12.9   Tue Oct 1 20:34:12 PDT 2024
```

OVERALL COMPARISON RESULTS

```
#####
#          #
#          #
#          # CORRECT #
#          #
#          #
#          #
#####
```



Simulation Waveform:

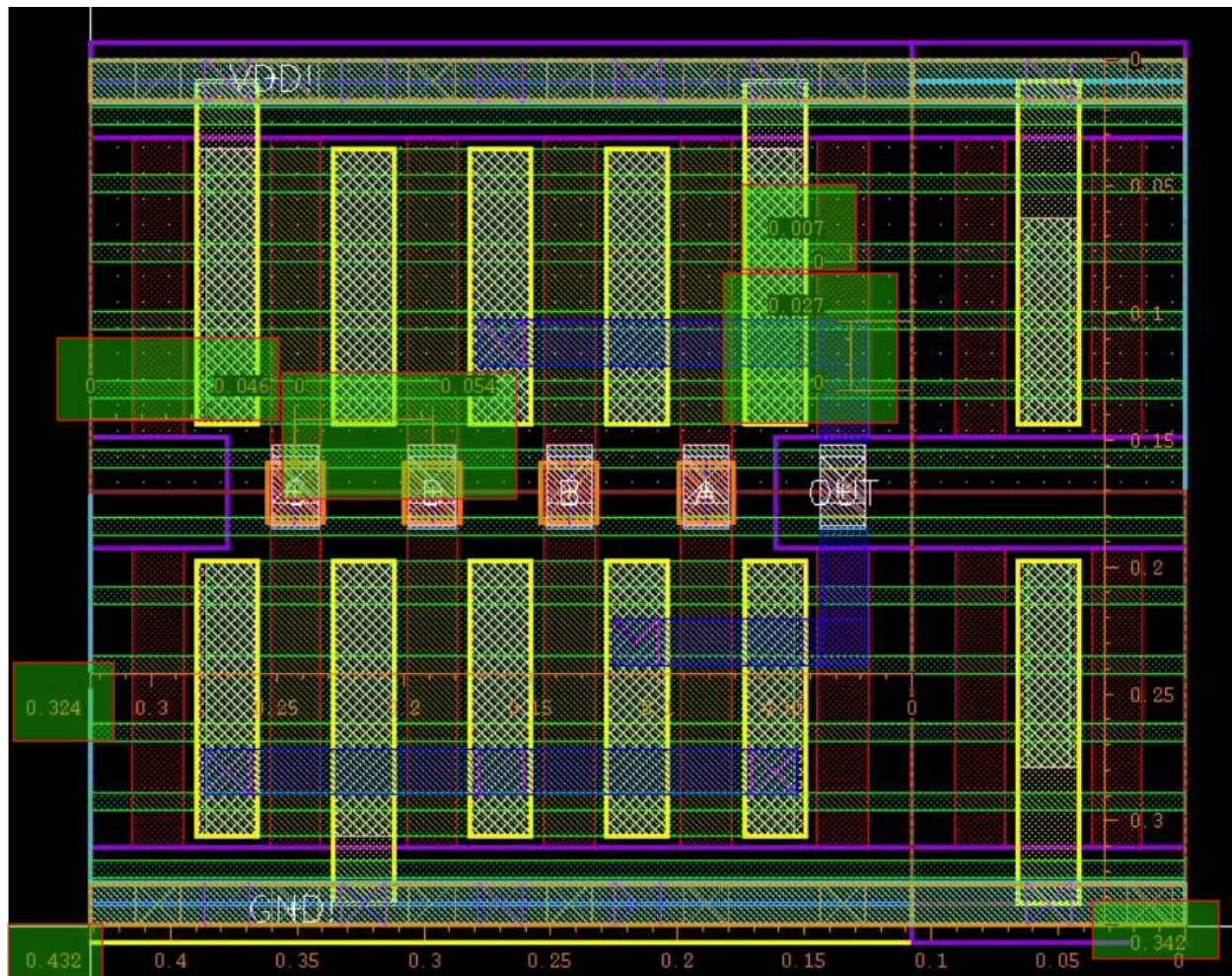


OAI22:

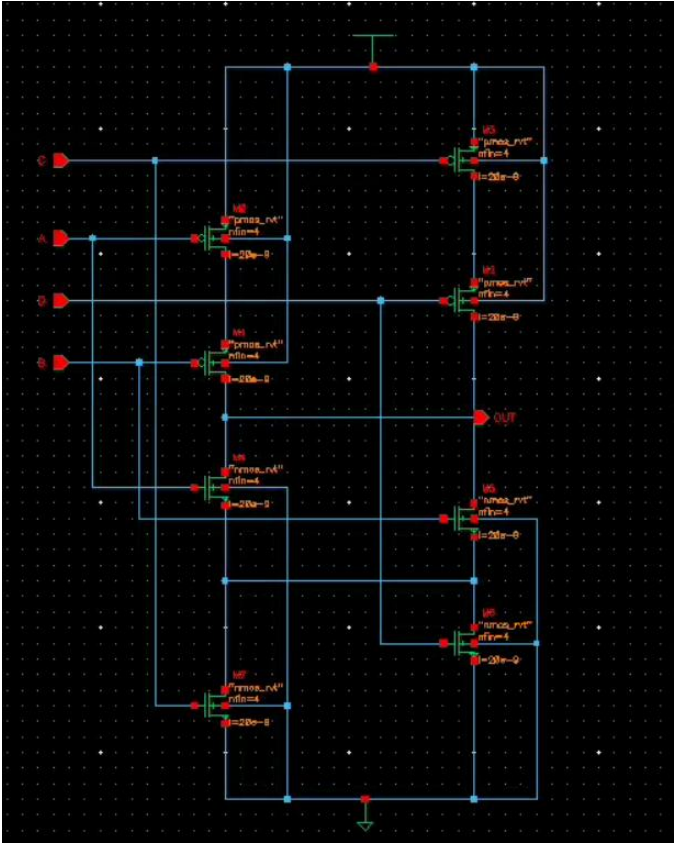
Layout design

The following Specs are used for the design

- Cell Height (VDD to GND) = 0.342um
- Cell width (with tap cell) = 0.432um
- Cell width (without tap cell) = 0.324um
- Cell offset = 0.046um
- Fin Pitch = 0.027um
- Fin width = 0.007um
- Gate Pitch (Pin to Pin Distance) = 0.054um



Schematic:



LVS Result:

CALIBRE SYSTEM ##
LVS REPORT ##

#####

REPORT FILE NAME: OAI22.lvs.report
LAYOUT NAME: OAI22.sp ('OAI22')
SOURCE NAME: OAI22.src.net ('OAI22')

RULE FILE: _calibre_LVS_rules_
RULE FILE TITLE: LVS Rule File for ASAP7 PDK
CREATION TIME: Wed Mar 26 00:23:10 2025
CURRENT DIRECTORY: /home/eng/m/mxi240020/7325/asap7/Std_cells/OAI22/LVS
USER NAME: mxi240020
CALIBRE VERSION: v2024.4_12.9 Tue Oct 1 20:34:12 PDT 2024

OVERALL COMPARISON RESULTS

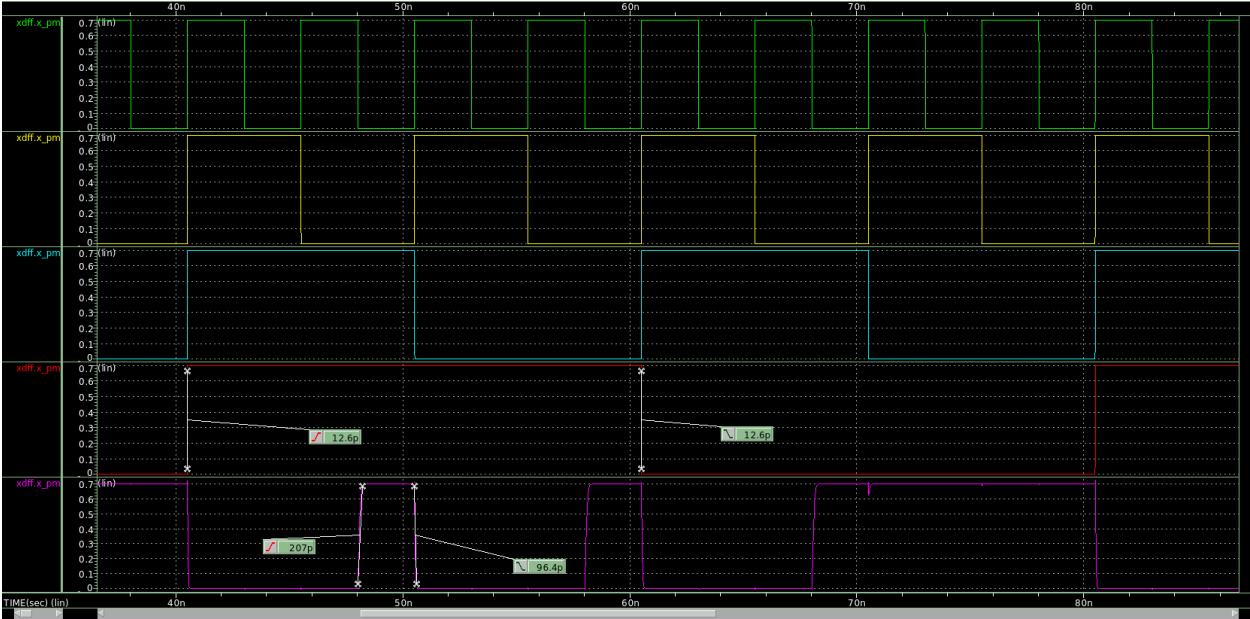
#

CORRECT #

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Simulation Waveform:

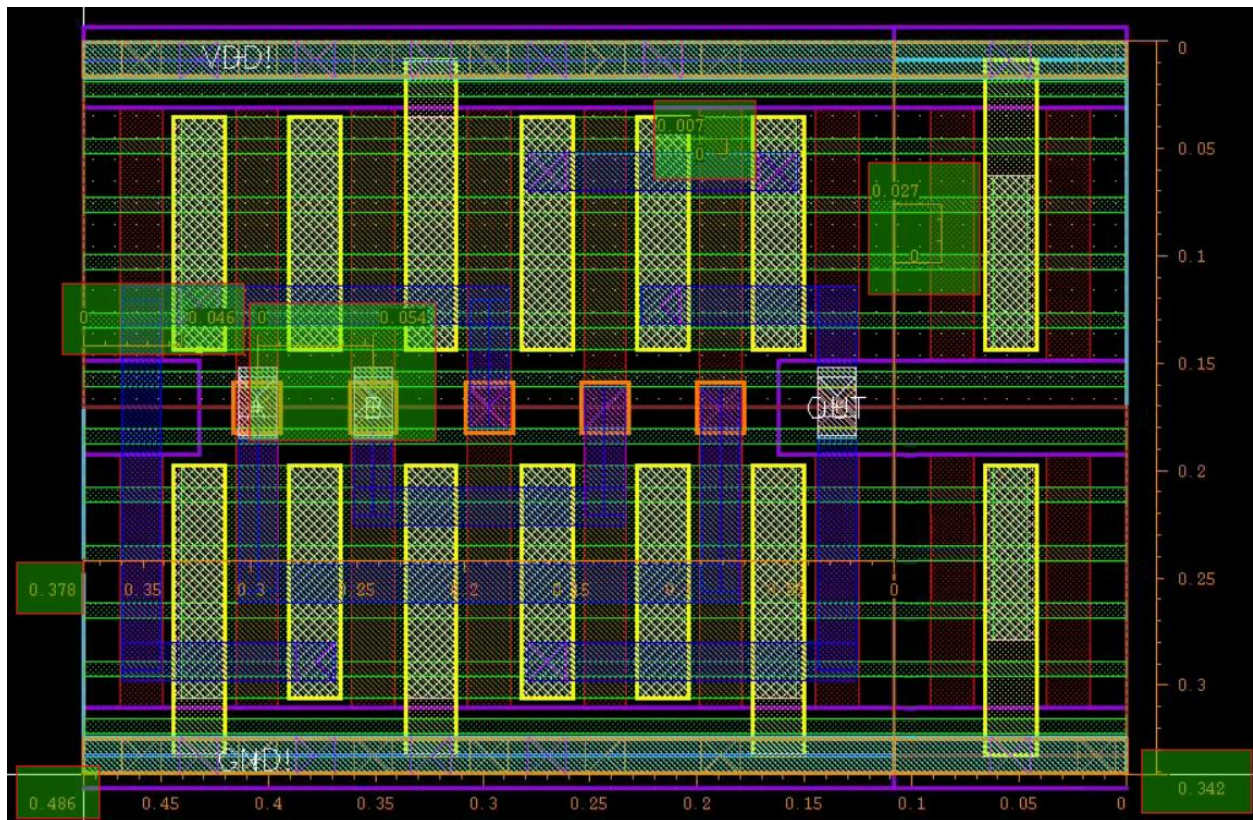


XOR:

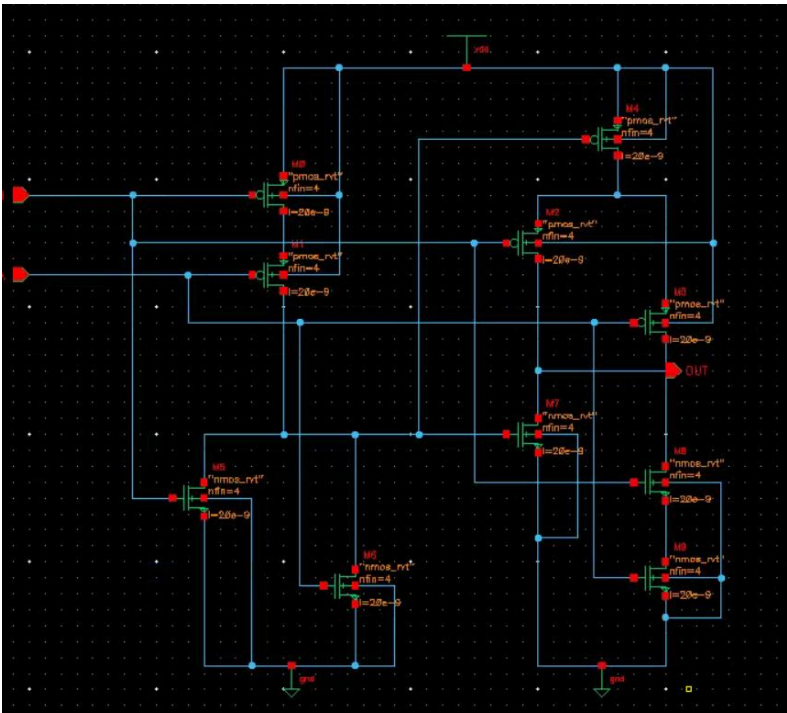
Layout design

The following Specs are used for the design

- Cell Height (VDD to GND) = 0.342um
- Cell width (with tap cell) = 0.486um
- Cell width (without tap cell) = 0.378um
- Cell offset = 0.046um
- Fin Pitch = 0.027um
- Fin width = 0.007um
- Gate Pitch (Pin to Pin Distance) = 0.054um



Schematic:



LVS Result:

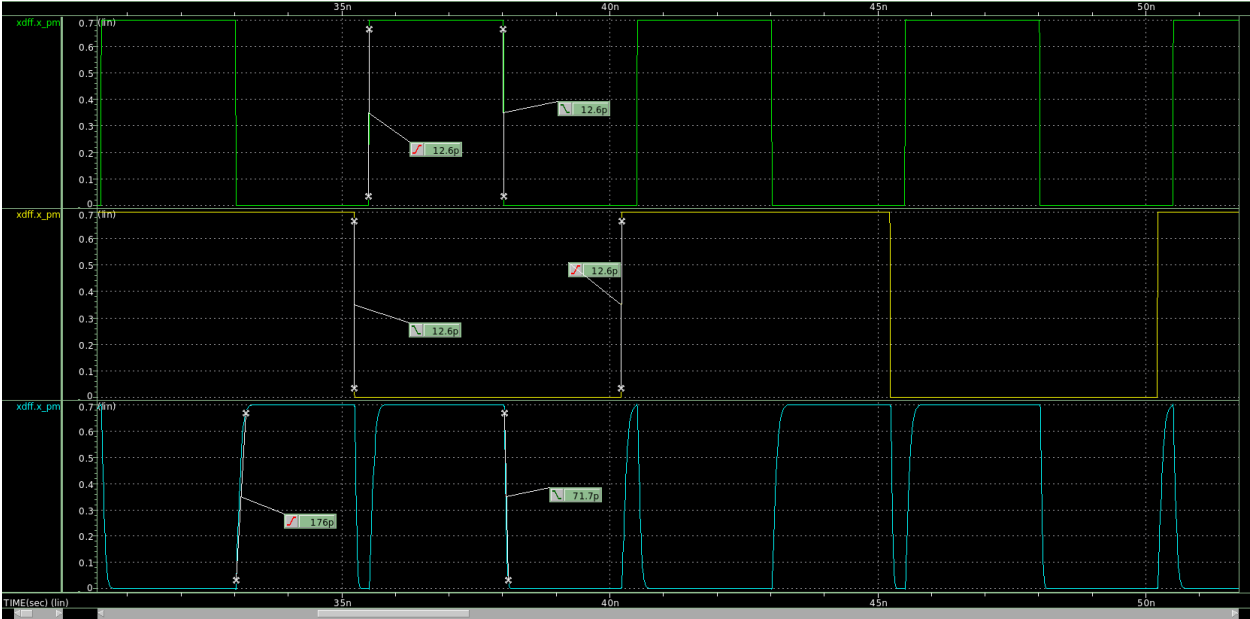
CALIBRE SYSTEM ##
LVS REPORT ##
#####

REPORT FILE NAME: XOR.lvs.report
LAYOUT NAME: XOR.sp ('XOR')
SOURCE NAME: XOR.src.net ('XOR')
RULE FILE: _Calibre_LVS_rules_
RULE FILE TITLE: LVS Rule File for ASAP7 PDK
CREATION TIME: Wed Mar 26 00:27:00 2025
CURRENT DIRECTORY: /home/eng/m/mx1240020/7325/asap7/Std_cells/XOR/LVS
USER NAME: mx1240020
CALIBRE VERSION: v2024.4_12.9 Tue Oct 1 20:34:12 PDT 2024

OVERALL COMPARISON RESULTS

CORRECT #
#####

Simulation Waveform:

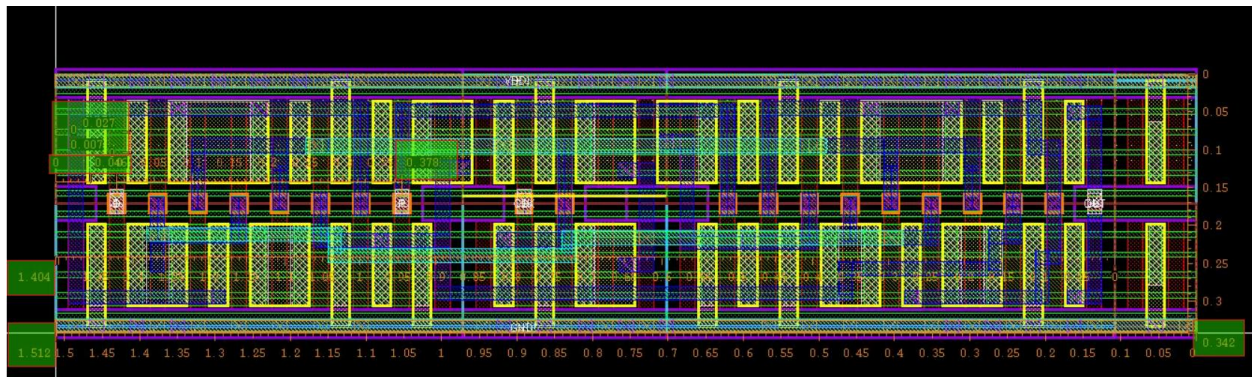


DFF:

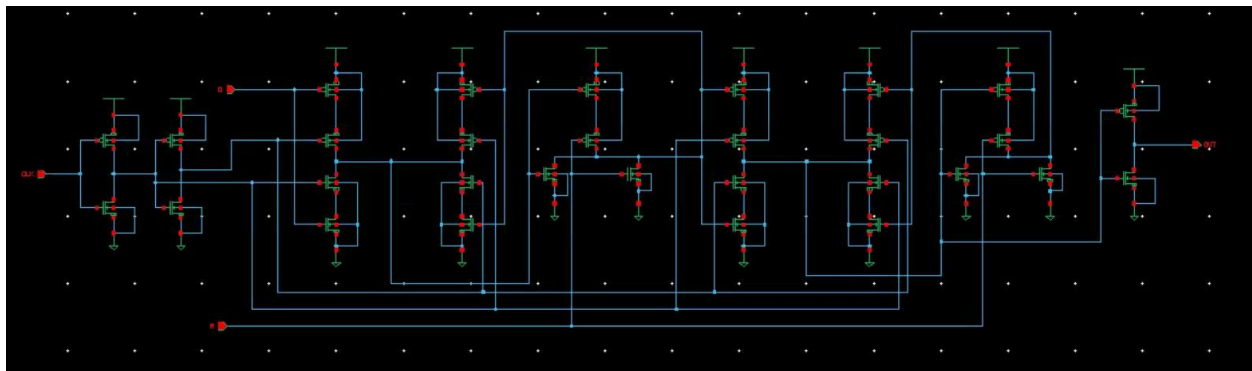
Layout design

The following Specs are used for the design

- Cell Height (VDD to GND) = 0.342um
- Cell width (with tap cell) = 1.512um
- Cell width (without tap cell) = 1.404um
- Cell offset = 0.046um
- Fin Pitch = 0.027um
- Fin width = 0.007um
- Gate Pitch (Pin to Pin Distance) = 0.378um (7 x 0.054um therefore it is multiples of the specified pitch)



Schematic:




```
#####  
##                                ##  
##          C A L I B R E    S Y S T E M      ##  
##                                ##  
##          L V S    R E P O R T              ##  
##                                ##  
#####
```

REPORT FILE NAME: DFF.lvs.report
LAYOUT NAME: DFF.sp ('DFF')
SOURCE_NAME: DFF.src.net ('DFF')

RULE FILE: _Calibre_LVS_rules_
RULE FILE TITLE: LVS Rule File for ASAP7 PDK
CREATION TIME: Thu Mar 27 02:42:27 2025
CURRENT DIRECTORY: /home/eng/m/mxi240020/7325/asap7/Std_cells/DFF/LVS
USER NAME: mxi240020
CALIBRE VERSION: v2024.4_12.9 Tue Oct 1 20:34:12 PDT 2024

OVERALL COMPARISON RESULTS

#

CORRECT #

#####

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HSPICE code

- Specifications
- Cout (Output Capacitance): 10fF
- Vdd (voltage between VDD and GND) = 0.7v
- SR (Slew Rate) = 14ps

1_input HSPICE code

```
1 $ HSPICE setup file for INV
2 $ Name: Lamin Jammeh & Mark Ivey
3 $ UTD_ID: dal852207 MXI240020
4 $ CE7325 Project1
5
6 $ transistor model this includes files generated from cadence virtuoso with the run.pex
7
8 .include "INV.pex.sp"
9
10 .option post runlvl=5
11
12 $ define the name for the circuit under test and ports
13 x1 GND! OUT VDD! IN INV
14
15 $define Power Supply as Vdd = 0.7v
16 Vdd VDD! GND! 0.7v
17
18 $ define Simulation parameters
19
20 .PARAM SR = 14ps $ Slew Rate
21 .PARAM T1 = 0ns $ Anticipated starting point
22 .PARAM T2 = 1ns
23 .PARAM T3 = 'T2 + SR' $ Ramp down using Slew rate
24 .PARAM T4 = 'T3 + T2'
25 .PARAM T5 = 'T4 + SR'
26 .PARAM T6 = 'T5 + T2'
27
28 $define an input signal for the simulation
29 VIN IN 0 PWL (T1 Vdd T2 Vdd T3 0v T4 0v T5 Vdd T6 Vdd)
30
31 $ Load capacitance = 10fFarad
32 Cout OUT GND! 10f
33
34 $ Type of HSPICE Simulation = Transient Analysis with 1ps step for 3000ps duration
35 .tran 1ps 3000ps
36
37
38 .end
```

2_input HSPICE code

```
1 $ HSPICE setup file for INV
2 $ Name: Lamin Jammeh & Mark Ivey
3 $ UTD_ID: dal852207 MXI240020
4 $ CE7325 Project1
5
6 $ transistor model this includes files generated from cadence virtuoso with the run.pex
7
8 .include "NAND2.pex.sp"
9
10 .option post runlvl=5
11
12 $ define the name for the circuit under test and ports
13 xi OUT GND! VDD! A B NAND2
14
15 $define Power Supply as Vdd = 0.7
16 Vdd VDD! GND! 0.7v
17
18 $ define Simulation parameters
19
20 .PARAM SR = 14ps $ Slew Rate
21 .PARAM T1 = 0ns $ Anticipated starting point
22 .PARAM T2 = 1ns
23 .PARAM T3 = 'T2 + SR' $ Ramp down using Slew rate
24 .PARAM T4 = 'T3 + T2'
25 .PARAM T5 = 'T4 + SR'
26 .PARAM T6 = 'T5 + T2'
27
28 $define the input signals A and B for the simulation using the possible logic combinations [(0,0) (0,1) (1,0) (1,1)]
29 VIN1 A 0 PWL (T1 0v T2 0v T3 0v T4 0v T5 Vdd T6 Vdd T7 Vdd T8 Vdd)
30 VIN2 B 0 PWL (T1 0v T2 0v T3 Vdd T4 Vdd T5 0v T6 0v T7 Vdd T8 Vdd)
31
32 $ Load capacitance = 10fFarad
33 Cout OUT GND! 10f
34
35 $ Type of HSPICE Simulation = Transient Analysis with 1ps step for 3000ps duration
36 .tran 1ps 3000ps
37
38 .end
```

3_input HSPICE code

```
1 $ HSPICE setup file for INV
2 $ Name: Lamin Jammeh & Mark Ivey
3 $ UTD_ID: dal852207 MXI240020
4 $ CE7325 Project1
5
6 $ transistor model this includes files generated from cadence virtuoso with the run.pex
7
8 .include "NAND3.pex.sp"
9
10 .option post runlvl=5
11
12 $ define the name for the circuit under test and ports
13 xi OUT GND! VDD! A B C NAND3
14
15 $define Power Supply as Vdd = 0.7
16 Vdd VDD! GND! 0.7v
17
18 $ define Simulation parameters
19
20 .PARAM SR = 14ps $ Slew Rate
21 .PARAM T1 = 0ns $ Anticipated starting point
22 .PARAM T2 = 2ns
23 .PARAM T3 = 'T2 + SR' $ Ramp down using Slew rate
24 .PARAM T4 = 'T3 + T2' $ first hold time
25
26 .PARAM T5 = 'T4 + SR' $ Ramp up using slew rate
27 .PARAM T6 = 'T5 + T2' $ second hold time
28
29 .PARAM T7 = 'T6 + SR' $ Ramp down using Slew rate
30 .PARAM T8 = 'T7 + T2' $ third hold time
31
32 .PARAM T9 = 'T8 + SR' $ Ramp up using slew rate
33 .PARAM T10 = 'T9 + T2' $ fourth hold time
34
35 $define the input signals A, B and C for the simulation using the possible logic combinations [(0,0,0) (0,0,1) (0,1,0) (0,1,1) (1,0,0) (1,0,1) (1,1,0) (1,1,1)]
36 VIN1 A 0 PWL (T1 0v T2 0v T3 0v T4 0v T5 Vdd T6 Vdd T7 Vdd T8 Vdd T9 Vdd T10 Vdd)
37
38 VIN2 B 0 PWL (T1 0v T2 0v T3 0v T4 0v T5 Vdd T6 Vdd T7 0v T8 0v T9 Vdd T10 Vdd)
39
40 VIN3 C 0 PWL (T1 0v T2 0v T3 Vdd T4 Vdd T5 0v T6 0v T7 Vdd T8 Vdd T9 Vdd T10 Vdd)
41
42 $ Load capacitance = 10fFarad
43 Cout OUT GND! 10f
44
45 $ Type of HSPICE Simulation = Transient Analysis with 1ps step for 3000ps duration
46 .tran 1ps 3000ps
47
48 .end
49
```

4_input HSPICE code

```
1 $ HSPICE setup file for INV
2 $ Name: Lamin Jamneh & Mark Ivey
3 $ UTD_ID: dal852207 MKI240020
4 $ CE7325 Project1
5
6 $ transistor model this includes files generated from cadence virtuoso with the run.pex
7
8 .include "NAND4.pex.sp"
9
10 .option post runlvl=5
11
12 $ define the name for the circuit under test and ports
13 xi OUT GND! VDD! A B C D NAND4
14
15 $define Power Supply as Vdd = 0.7
16 Vdd VDD! GND! 0.7v
17
18 $ define Simulation parameters
19
20 .PARAM SR = 14ps $ Slew Rate
21 .PARAM T1 = 0ns $ Anticipated starting point
22 .PARAM T2 = 1ns $ Ramp down using Slew rate
23 .PARAM T3 = 'T2 + SR' $ first hold time
24 .PARAM T4 = 'T3 + T2' $ Ramp up using slew rate
25 .PARAM T5 = 'T4 + SR' $ second hold time
26 .PARAM T6 = 'T5 + T2' $ Ramp down using Slew rate
27 .PARAM T7 = 'T6 + SR' $ third hold time
28 .PARAM T8 = 'T7 + T2' $ Ramp up using slew rate
29 .PARAM T9 = 'T8 + SR' $ fourth hold time
30 .PARAM T10 = 'T9 + T2' $ Ramp up using slew rate
31 .PARAM T11 = 'T10 + SR' $ second hold time
32 .PARAM T12 = 'T11 + T2' $ Ramp down using Slew rate
33 .PARAM T13 = 'T12 + SR' $ third hold time
34 .PARAM T14 = 'T13 + T2' $ Ramp up using slew rate
35 .PARAM T15 = 'T14 + SR' $ fourth hold time
36 .PARAM T16 = 'T15 + T2'
37
38 $define the input signals VA, VB and C for the simulation usig the possible logic combinations [(0,0,0,0) (0,0,1,0) (0,1,0,1) (0,1,1,0) (1,0,0,1) (1,0,1,1) (1,1,0,1) (1,1,1,1)]
39 VIN1 A 0 PWL (T1 0v T2 0v T3 0v T4 0v T5 0v T6 0v T7 0v T8 0v T9 Vdd T10 Vdd T11 Vdd T12 Vdd T13 Vdd T14 Vdd T15 Vdd T16 Vdd)
40
41 VIN2 B 0 PWL (T1 0v T2 0v T3 0v T4 0v T5 Vdd T6 Vdd T7 Vdd T8 Vdd T9 0v T10 0v T11 0v T12 0v T13 Vdd T14 Vdd T15 Vdd T16 Vdd)
42
43 VIN3 C 0 PWL (T1 0v T2 0v T3 Vdd T4 Vdd T5 0v T6 0v T7 Vdd T8 Vdd T9 0v T10 0v T11 Vdd T12 Vdd T13 0v T14 0v T15 Vdd T16 Vdd)
44
45 VIN4 D 0 PWL (T1 0v T2 0v T3 0v T4 0v T5 Vdd T6 Vdd T7 0v T8 0v T9 Vdd T10 Vdd T11 Vdd T12 Vdd T13 Vdd T14 Vdd T15 Vdd T16 Vdd)
46
47 $ Load capacitance = 10fFarad
48 Cout OUT GND! 10f
49
50 $ Type of HSPICE Simulation = Transient Analysis with 1ps step for 3000ps duration
51 .tran 1ps 3000ps
52
53 .end
54
55
56
57
58
59
60
61
```